

STRESS AND STRAIN IN ISOTROPIC MATERIALS

Section: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3101**

Title: **STRESS AND STRAIN IN ISOTROPIC MATERIALS**

Revision: **1.2**

Date: **22/05/2023**

1 INTRODUCTION

1.1 OBJECT

Classical theory of Elasticity (Hook's law) can be applied in the linear elastic range of isotropic metal alloys. Classical theories about elasticity and stress analysis shall not be treated here exhaustively. Suggested bibliography to review, among others, in the list of reference of appendix.

The typical expression of the fundamental relationships between stress and strain, though material engineering constants (Young modulus and Poisson's ratio), in a bi-dimensional stress state:

$$\varepsilon_x = \sigma_x/E - \nu \cdot \sigma_y/E \quad \varepsilon_y = \sigma_y/E - \nu \cdot \sigma_x/E \quad \gamma_{xy} = \tau_{xy}/G$$

Mohr's circle and related expressions to obtain the principal stresses of a bi-axial stress state are also worthy to remind here:

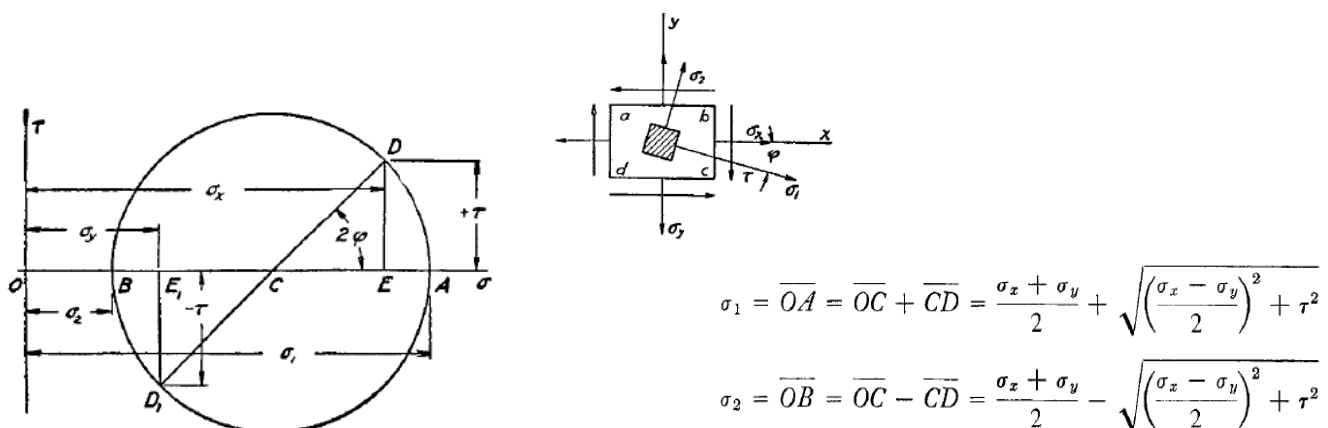


Figure 1. Mohr's circle formulas for principal stresses, from Timoshenko

STRESS AND STRAIN IN ISOTROPIC MATERIALS

The inelastic behavior will be modelled using the well known Ramberg-Osgood approximation, using the engineering values for stress and strain, except for 3D-DFEMs where real values are typically recommended. Relationship between engineering and real values for stress and strain:

$$\sigma_{true} = \sigma(1+\epsilon) \quad \epsilon_{true} = \ln(1+\epsilon)$$

Engineering stress and strain

$$\sigma = F / A_0$$

$$\epsilon = \Delta L / L_0$$

True stress and strain

$$\sigma_{true} = F / A = (F / A_0) (A_0 / A) = (F / A_0) (L / L_0) = \sigma(1+\epsilon)$$

$$\epsilon_{true} = \int_{L_0}^L \partial L / L = \ln(L_0 + \Delta L) - \ln(L_0) = \ln(1 + \Delta L / L_0) = \ln(1 + \epsilon)$$

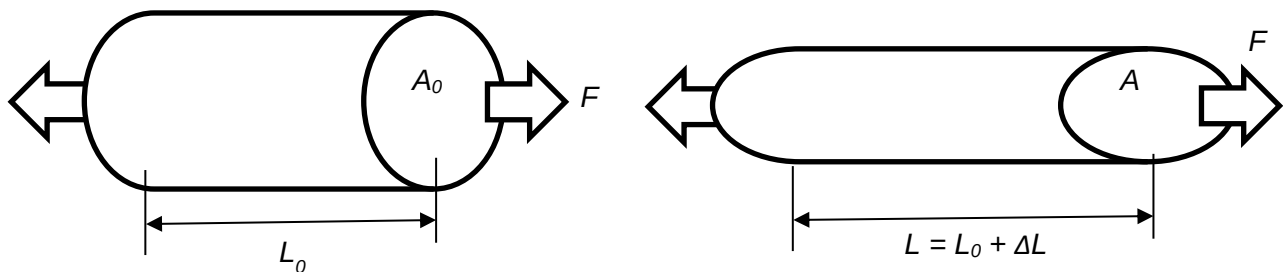


Figure 2. Engineering and true stresses and strain in a tension bar

The Ramberg-Osgood equation:

$$\epsilon = \sigma/E + 0.002(\sigma/F_y)^n$$

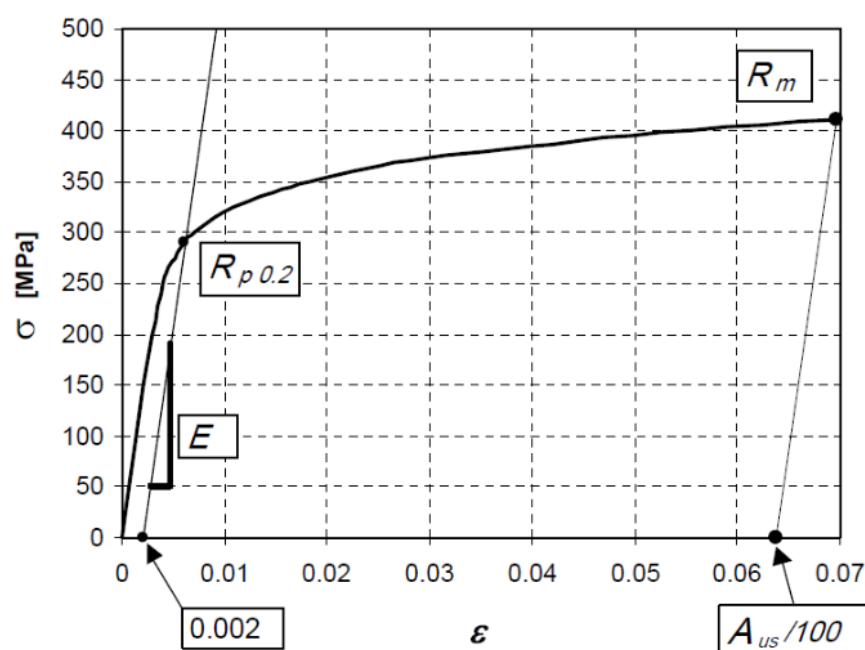


Figure 3. Example of stress-strain curve per Ramberg-Osgood model (from HSB 52110-01)

STRESS AND STRAIN IN ISOTROPIC MATERIALS

However, the Ramberg-Osgood model is valid only for describing the stress-strain relationship at load levels below the ultimate tensile stress of the material. Beyond F_{tu} , a metal deforming plastically under a tensile load may experience highly localized extension and thinning, called necking, as the material fails. Even classical methodologies for analyzing stress and safety margins in metallic materials match well with the Ramberg-Osgood model and this limitation, the development of novel numerical high-fidelity methods for predicting accurately failure loads in structures have generated the need of modelling the material behavior also after this limit. Theories about Continuum Damage Mechanics and its implementations are examples of these new methods. In any case, this topic is considered out of the scope of this chapter and shall not be discussed.

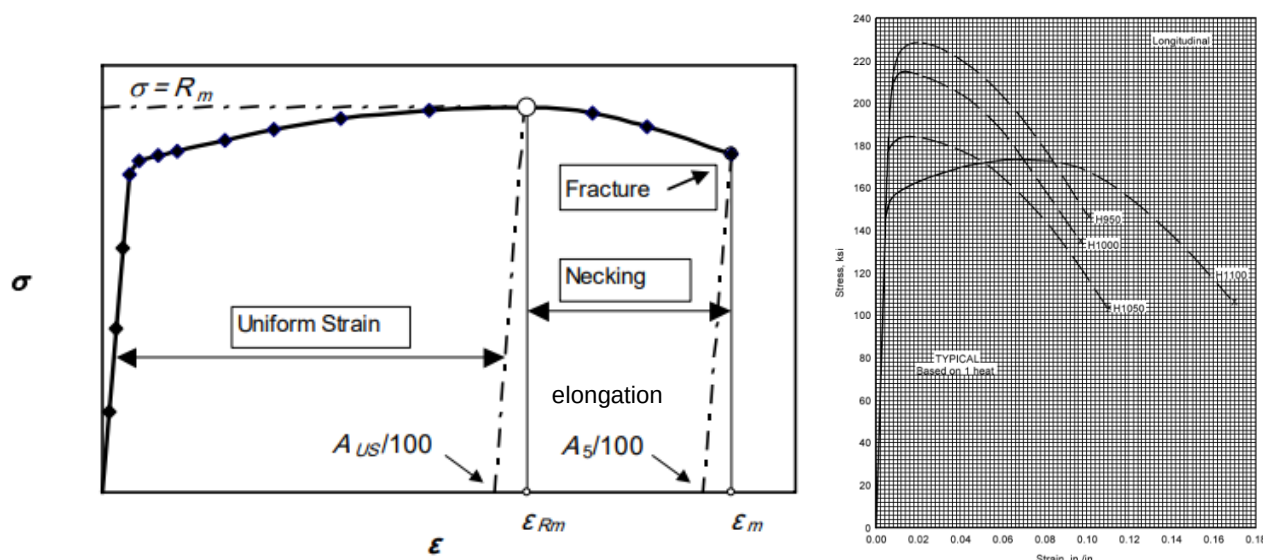


Figure 4. Full-range stress-strain curve: sketch and curves for PH13-8Mo (HSB 52110-01)

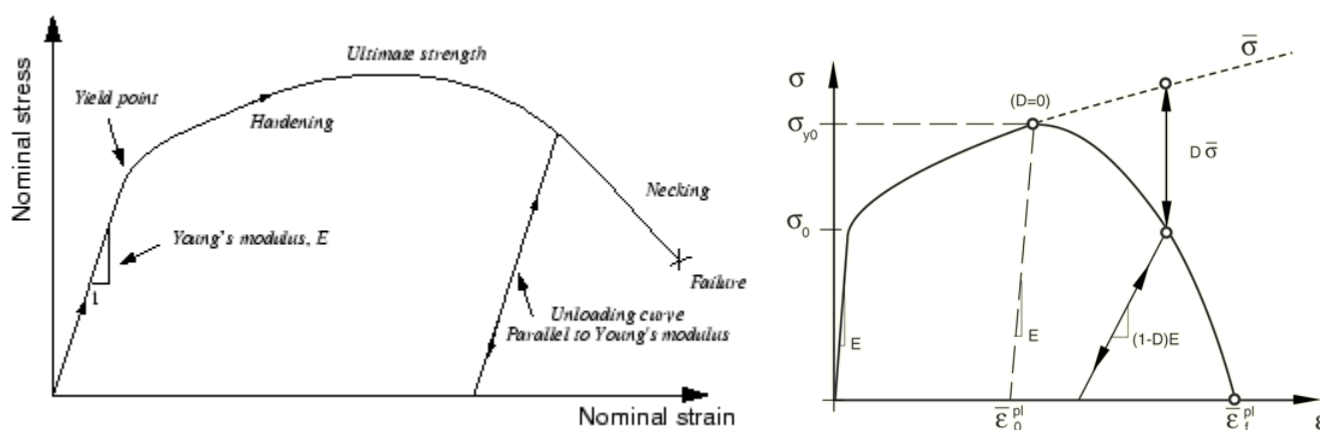


Figure 5. Schemes of stress-strain curves in user documentation of software Simulia Abaqus

For some particular design features and failure modes the plastic correction may be treated with dedicated methods (without explicitly modelling the stress-strain curve). See chapters dedicated to local buckling, beams bending, ...

STRESS AND STRAIN IN ISOTROPIC MATERIALS

2 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
e	%	Elongation, i.e. permanent deformation after failure, typically in % ($A_{us}/100$ in HSB sheets)
E	MPa	Young's modulus
E_c	MPa	Elastic modulus in compression
ϵ	-	Axial strain
F_{cy}	MPa	Yield compression stress
F_{ty}	MPa	Yield tensile stress (Rp0.2 in HSB sheets)
UTS, F_{tu}	MPa	Ultimate tensile stress (Rm in HSB sheets)
F_y	MPa	Yield stress (either tension or compression)
G	MPa	Shear elastic modulus, in isotropic materials: $G = E / [2(1+\nu)]$
γ	MPa	Shear strain
n	-	Ramberg-Osgood exponent
ν	-	Poisson's ratio
σ	MPa	Axial stress
τ	MPa	Shear stress

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[S. Timoshenko]	3 rd	STRENGTH OF MATERIALS
Ref. 2	[W. C. Young & R. G. Budynas]	7 th	ROARKS'S FORMULAS FOR STRESS AND STRAIN
Ref. 3	[S. Timoshenko et al.]	2 nd	THEORY OF PLATES AND SHELLS
Ref. 4	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 5	[M. C. Y. Niu]	2 nd	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 6	HSB 52110-01	Is. B	The Ramberg-Osgood stress-strain relationship for metallic materials
Ref. 7	[SAE] MMPDS	9	Metallic Materials Properties Development and Standardization
Ref. 8	[Dassault Systèmes]	-	Abaqus Analysis User's Guide
Ref. 9	HSB 12522-01	Is. C	Ramberg-Osgood exponents of aeronautical metallic materials

STRESS AND STRAIN IN ISOTROPIC MATERIALS

SOFTWARE

N/A

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 01/08/2018	G. Baños	First issue	All pages
1.1 26/04/2023	G. Baños	Remove expression relating the Ramberg-Osgood exponent and the elongation after failure. Add mention to the limitation of this model after the material <i>UTS</i> .	Pages 2 - 3
1.2 22/05/2024	G. Baños	Explanations about definitions and relationships for engineering and true stresses and strains	Page 2

Table 1. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis	-	Florian Glock 23/05/2024
Stress Methods	G. Baños 22/05/2024	-

Table 2. *Approval signatures table for last revision of this chapter*